MIL-D-40123C
14 June 1983
SUPERSEDING
MIL-D-40123B
5 September 1968

MILITARY SPECIFICATION

DOUGH TROUGH, FERMENTATION, FIELD

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document covers one type of dough fermentation trough with casters.

2. APPLICABLE DOCUMENTS

2.1 Government documents. Unless otherwise specified, the following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this document to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-S-681	-	Steel Castings
TT-E-529	-	Enamel Alkyd, Semi-Gloss
TT-P-1757	-	Primer Coating, Zinc Chromate, Low Moisture Sensitivity
MMM-A-1617	-	Adhesive, Rubber Base, General Purpose

MILITARY

MIL-B-2316	-	Bakery Equipment, Including Units, Assemblies, Repair Parts and Tools, Preparation for Delivery of
MIL-C-5541	-	Chemical Conversion Castings For Aluminum and Aluminum Alloys

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Natick Research and Development Laboratories, Natick, MA 01760 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 7320

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STANDARDS

FEDERAL

FED-STD-595 - Colors

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

DRAWINGS

US ARMY NATICK RESEARCH AND DEVELOPMENT LABORATORIES

5-11-784	-	Dough Trough, Fermentation, Field; Assembly
5-11-785	-	Dough Trough, Fermentation, Field; Cover Assembly
5-11-786	-	Dough Trough, Fermentation, Field; Panels
5-11-787	-	Dough Trough, Fermentation, Field, Dam Board Assembly
5-11-788	-	Dough Trough, Fermentation, Field; Details
5-11-789	-	Dough Trough, Fermentation, Field; Details

(Copies of documents required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order or precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document shall take precedence.

3. REQUIREMENTS

* 3.1 First article. When specified, a sample shall be subjected to first article inspection (see 4.3, 6.2, and 6.3).

3.2 Materials (see 6.4) and components. The materials and components shall be as specified herein or as specified on applicable drawings.

3.2.1 Hardware. All hardware shall be of the design, type, and material as shown on applicable drawings, except the handle bracket, item 9 on Drawing 5-11-784 and 5-11-789, may be cast from material conforming to classes 65-35 or 70-36 of QQ-S-681.

3.2.2 Solder. Solder shall conform to composition Sn 96 of QQ-S-571.

3.2.3 Adhesive. Adhesive used for cementing the insulating board to the corrosion-resisting steel and aluminum sheets shall conform to type II of MMM-A-1617.

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3.2.4 Primer. Zinc chromate primer shall conform to TT-P-1757.

3.2.5 Enamel. Enamel shall conform to TT-E-529. Color shall be Olive Drab No. X24087 conforming to FED-STD-595.

3.3 Construction. The construction of the dough trough shall conform to the requirements herein and the design, dimensions, material classifications, and sizes thereof shown on drawings listed in 2.1.

3.3.1 Body and cover. The body and cover of the dough trough shall be panel construction having an inner liner of stainless steel, an insulated core and an outer liner of aluminum. The insulation shall be bonded to the panels with adhesive specified in 3.2.3. The stainless steel, aluminum sheet, and insulation prior to bonding, shall be thoroughly cleaned with a suitable solvent, such as toluene. All inside seams shall be filled with solder specified in 3.2.2. The body handles shall be attached as shown on Drawing 5-11-789.

3.3.2 Board, dam. The dam board shall be paneled and insulated the same as the dough trough body and cover. After fabrication, the dam board shall fit freely in the runners in the location shown on Drawing 5-11-784.

3.4 Finish.

3.4.1 Surface preparation. Prior to assembly, all aluminum parts shall be given a treatment conforming to class 2 of MIL-C-5541. All surfaces of steel and aluminum, except aluminum rivets, which are in direct contact with each other, shall be covered with one coat of primer specified in 3.2.4. All stainless steel rivets shall be given a coat of primer specified in 3.2.4.

3.4.2 Cleaning. After assembly of the trough, all exterior surfaces shall be thoroughly cleaned and degreased followed by a thorough drying. Special attention shall be given to drying out crevices and other places difficult to dry.

3.4.3 Priming. All exterior surfaces of the cover and body shall be given a coat of zinc chromate primer specified in 3.2.4 with a dry film thickness of 0.4 mil minimum.

3.4.4 Painting. When thoroughly dry, the primer shall be followed by two coats of enamel specified in 3.2.5 with a dry film thickness of 1.6 mil minimum. The combined thickness shall level out to a smooth and uniform film without application marks, orange peel, runs, wrinkles, drops, or areas of thin film or no film, and shall be free of grit or coarse particles. The primer and enamel coat shall be free of blisters. All surfaces of the trough and cover which will be in direct contact with the dough shall remain unfinished.

3.5 Marking for identification. The trough shall be marked by indenting of the letters "U.S.", the manufacturer's name, trade name or trademark and date of the manufacture of the size, and in the location shown on Drawing 5-11-784.

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3.6 Workmanship. Components shall be assembled in a neat manner. The dam board shall not bind. Painted areas shall be smooth, uniformly coated, free of runs, wrinkles, orange peel, grit, or areas of no film.

3.6.1 Metal fabrication. All metal used in the fabrication of the dough trough shall be free of kinks, sharp bends, and slivers. Shearing shall be done neatly and accurately. Bends shall be made accurately to provide uniformity of size and shape. All pins, studs, and screw holes shall be accurately drilled and all burrs removed.

3.6.2 Riveting. Holes for rivets shall be accurately punched or drilled. Rivets shall be tight, fully peened, and heads in full contact with the surface of the metal. All rivets where possible shall be driven from inside of the dough trough and peened on the outside. Stems of all exposed rivets shall be ground smooth.

3.6.3 Soldering. All inside seams shall be soldered and have a fillet of not less than 1/4 inch radius and shall be finished smooth with no holes, fissures, fins, or rough spots.

3.6.4 Welding. Surfaces of all parts to be welded shall have oxide, scale, paint, grease and other foreign matter removed. The welds shall be continuous, adherent, smooth and free of cracks, pits, imbedded foreign material, flux, and scale.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the document where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

4.3 First article inspection. When a first article is required (see 6.2), it shall be examined for the defects specified in table II and 4.4.5. The presence of any defect shall be cause for rejection of the first article.

4.4 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

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4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified or qualified in this document or applicable purchase document.

4.4.2 Operation examination. Examination shall be made during the manufacturing to insure that the operations listed below are performed:

(a) Aluminum surfaces are treated (see 3.4.1).
(b) Corrosion-resisting steel sheets, aluminum sheet and insulation are cleaned prior to bonding (see 3.3.1).
(c) Aluminum and steel surfaces which are in direct contact are coated with primer (see 3.4.1).
(d) Assembled dough trough is thoroughly cleaned, degreased and dried prior to application of finish (see 3.4.2).
(e) Primer is applied to the exterior surfaces prior to painting (see 3.4.3).

Whenever nonconformance is noted, correction or replacement shall be made to affected items and process.

4.4.3 In-process examination. Prior to sealing, the panels shall be examined for defects listed in table I. The lot size shall be expressed in units of panels. The sample unit shall consist of one side, end, bottom, left cover or right cover panel, prior to sealing of each panel. The inspection level shall be level II, and the AQL, expressed in terms of defects per hundred units, shall be 6.5.

TABLE I. Panel insulation defects

Examine	Defect	Classification Major	Minor
Insulation	Omitted, not type specified.	X	
	Not butted tight against panel frame tubing.	X	
	Evidence that rubber is not firmly cemented with specified adhesive to both panels and tubing.		X
	Crumbled or deteriorated.	X	
	Evidence that scraps or waste materials are being used or voids in panel.	X	

4.4.4 End item visual examination. The end item shall be examined for the defects listed in table II. The lot size shall be expresssed in units of dough troughs. The sample unit shall be one dough trough. The inspection level shall be II and the acceptable quality level (AQL), expresssed in terms of defects per hundred units, shall be 2.5 for major defects and 6.5 for total (major and minor combined) defects.

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TABLE II. End item defects

Examine	Defect	Classification Major	Minor
Finish	Wrong color or type of finish; finish missing.	X	
	Blistered, peeled, chipped or area of not film or thin film.		X
	Separation of color, touch-up not neatly done, stained, overspray, foreign matter in finish, or finish otherwise impaired.		X
	Finish not adherent or is tacky.		X
	Surface discolored, scratched, wrinkled, rippled, orange peel, sags or runs.		X
	Corrosion-resistant steel not natural finish and clean.	X	
Construction and workmanship	Any part wrong type or design.	X	
	Any part broken, fractured, punctured, cracked, damaged, malformed, dented, buckled, warped or bent out of shape.	X	
	Any pronounced tool or forming marks.	X	
	Any part missing.	X	
	Parts misplaced, loose or not in proper alignment.	X	
	Any sharp edges, corners, burrs, slivers or projections.	X	
	Cover requires abnormal force to operate.		X
	Any component not connected or poorly joined as specified or poorly accomplished.	X	
	Any specified operation omitted.	X	
	Any weld missing, completely broken or not fused.	X	
	Weld not smooth, sound or free of flashes or splatter.	X	
	Any functional part that requires excessive force to operate.		X
	Parts not connected or joined as specified.	X	
	Soldered on joint or seams not finished smooth, solder missing on seams or joints not completely closed with solder where specified, not adherent or cracked.	X	

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TABLE II. End item defects (cont'd)

Examine	Defect	Classification Major	Minor
Construction and workmanship (cont'd)	Paint on interior surfaces.	X	
	Dirt or foreign matter inside compartment.		X
	Cover does not fit properly.	X	
	Handle assembly loose or not riveted as specified.	X	
	Handle assembly not installed as specified or does not function properly.	X	
	Dough drough not rigid or is out of alignment.	X	
	Casters do not swivel or do not function as intended.	X	
	Casters not tired.		X
	Any rivet not spaced as specified, or is improperly set (loose head cocked or not peened).	X	
	Rivet missing, wrong type or size.	X	
	Rivet not ground flush where required.	X	
	Omitted, incomplete, incorrect, illegible or not located properly.		X

4.4.5 Dimensional examination. Inspection shall be made of the dough trough for compliance with the dimensions specified on the drawings. Any deviation from the specified requirements shall constitute a defect. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 4.0.

4.4.6 Preparation for delivery inspection. Inspection of preparation for delivery shall be in accordance with the quality assurance provisions of MIL-B-2316.

5. PREPARATION FOR DELIVERY

5.1 Levels A, B, and C. Preservation, packaging, packing, and marking for each level of protection shall be in accordance with the applicable requirements of MIL-B-2316 (see 6.2).

6. NOTES

6.1 Intended use. The dough fermentation trough is intended for the fermentation of dough for field ration bread. The dough trough is a component of the Bakery Plant, Field, Portable, consisting of 170 components, MIL-B-43322.

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6.2 Sources of documents

6.2 Ordering data. Acquisition documents should specify the following:

(a) Title, number, and date of this document.
(b) When a first article is required (see 3.1, 4.3, and 6.3).
(c) Selection of the applicable levels of preservation, packaging, and packing (see 5.1).

* 6.3 First article. When a first article sample is required, is shall be inspected and approved under the appropriate provisions of DAR 7-104.55. The first article should be a preproduction sample consisting of one complete dough trough. The contracting officer should include specific instructions in all acquisition documents, regarding arrangements for inspection and approval of the first article.

6.4 Recycled material. It is encouraged that recycled material be used when practical as long as it meets the requirements of the document (see 3.2).

6.5 Changes from previous issue. The margins of this document are marked with an asterisk to indicate where changes (additions, modification, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-D-401230	Dough Trough, Fermentation, Field

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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